Project Management Plan & Risk Register: AayurLink

Project Parameters

- **Methodology:** Agile Scrum (1-Week Sprints over a 4-Week Fast-Track Lifecycle)
- **Project Manager:** [Your Name]
- **Tracking Tool:** Jira Software / Trello Kanban
- **Target Delivery:** Production MVP (Responsive Web Application)

1. Work Breakdown Structure (WBS) & Sprint Schedule

Sprint 1: Foundation, Auth & Data Schema (Week 1)

- **S1-T1:** Finalize Business Requirements Document (BRD), User Personas, and Process Flow Diagrams.
- **S1-T2:** Setup Git Repository and Project Scaffolding (Next.js / Tailwind CSS).
- **S1-T3:** Configure Database Schema (Supabase / Firebase) for Users, Slots, and Vitals.
- **S1-T4:** Implement US-01 (Phone OTP Mock Login with Sinhala/English language toggle).

Sprint 2: Core Dialysis Reservation Engine (Week 2)

- **S2-T1:** Design High-Fidelity UI Screens in Figma (Mobile and Desktop views).
- **S2-T2:** Build Slot Engine Database Tables (Hospitals, Shifts, Machines, Reservations).
- **S2-T3:** Implement US-02 (Slot selection, capacity validation, and digital token generation).
- **S2-T4:** Build automated scheduling rules (prevent duplicate bookings on identical dates).

Sprint 3: Clinical Vitals Logger & Water Directory (Week 3)

- **S3-T1:** Implement US-03 (Serum Creatinine and eGFR input form with clinical validation).
- **S3-T2:** Integrate dynamic line chart (Recharts / Chart.js) for longitudinal trends.
- **S3-T3:** Implement US-04 (RO Plant Directory list with district filter and operational status badge).
- **S3-T4:** Embed responsive map view with geolocation markers for active RO water points.

Sprint 4: Hospital Nurse Portal, UAT & Deployment (Week 4)

- **S4-T1:** Implement US-05 (Nurse Portal: Daily shift roster and 1-tap Check-in action).
- **S4-T2:** Conduct End-to-End User Acceptance Testing (UAT) across mobile viewports.
- **S4-T3:** Resolve UI/UX defects, optimize load times, and deploy production build to Vercel.
- **S4-T4:** Compile Project Portfolio Showcase (Case Study, Screenshots, and GitHub Documentation).

2. Comprehensive Risk Register & Mitigation Strategy

Risk ID	Risk Event Description	Severity	Likelihood	Mitigation Strategy (Action Plan)
R-01	Rural users find digital booking too complex (Low Tech Literacy)	High	High	Provide full Sinhala localization, high visual contrast, large touch targets, and surrogate caregiver profiles.
R-02	Clinic staff resist abandoning physical paper logbooks	High	Medium	Design a single-tap check-in UI that closely mirrors traditional paper register columns.
R-03	Project schedule slips past the 4-week fast-track window	Medium	Medium	Enforce strict MoSCoW prioritization; defer non-critical features (e.g., live SMS gateway) to Phase 2.

R-04	**Patient health data exposed without proper authorization**	**High**	**Low**	**Implement Row-Level Security (RLS) on the database, mask patient IDs, and store zero unencrypted PII.**
R-05	**Inaccurate RO plant data misleads patients to defunct water points**	**Medium**	**Medium**	**Include clear "Last Verified Date" stamp on every water point card and a community report flag button.**